

Mudassar Ali

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Summary

Mechanical Engineer with experience in manufacturing, CAD, and embeded electronics across different design, manufacturing and robotics projects.Skilled in evaluating required product demands and innovate, manufacture, prototype and design a working solution.

Education

NUST College Of Electrical and Mechanical Engineering

Bachelor of Mechanical Engineering,

Rawalpindi, Pakistan

Sep 2022- May 2026

Experience

National Centre of Robotics and Automation (NCRA)

Rawalpindi

Intern – Robot Design & Development Lab

Jun 2025–Jul 2025

- Contributed to the "Robodog" project, focusing on the integration of biomimetic movement and mechanical stability.
- Accelerated the design iteration cycle for structural components by leveraging rapid prototyping techniques, including 3D printing and laser cutting.
- Performed rigorous structural analysis in ANSYS and 3D modelling in SolidWorks to ensure the mechanical integrity and optimized motion of 4-bar linkage systems.

Research and Development Center (RDC), NUST

Rawalpindi

Project Contributor

Oct 2025–Apr 2025

- Streamlined manufacturing workflows through high-precision digital documentation and mechanical design.
- Comprehensive 2D and 3D CAD drawings for an electric bike project, facilitating seamless fabrication transitions and reducing production errors.
- Design and optimization of a hydraulic workbench, ensuring ergonomic efficiency, load-bearing reliability, and practical applicability.

Projects

Final Year Project

Sep 2025–May 2026

VTOL UAV

- Designed and developed a tricopter tilt roter VTOL drone equipped with onboard AI detection system.
- Integrated an open source AI vision system using Pi and camera to detect survivors during mass floods covering entire regions locating thousands of survivors and optimizing planning for rescue missions and AI usecase in interceptiing attack drones.

Skyscraper Cleaning Robot (Team Lead)

Nov 2024–Jan 2025

- Designed an automated robotic alternative to high-risk manual labor in high-rise maintenance.
- Developed a smart wire-grip movement system utilizing tilt correction and obstacle detection to ensure operational safety and efficiency without relying on conventional, hazardous open lifts.

Autonomous Solar Cleaning Robot (Ongoing)

Apr 2024–Present

- Assessed the feasibility of automated cleaning systems and designing a smart robot to address the flaws in current solutions.
- Integration of autonomous navigation and vision to clean multiple solar panels array of any layout without any supervision.

Achievements & Leadership

Electric Human-Powered Vehicle Competition (EHPVC)

Team Lead

Feb 2024–May 2025

Led 10-member team in the structural design, engineering, and successful fabrication of an electric human-powered vehicle for Pakistan's inaugural EHPV Competition.

Annual Machine Design Competition

3rd Position

2023

Designed and Manufactured a Hydraulic Arm recognized for excellence in mechanical conceptualization, problem-solving, and practical design execution.

Zindigi Prize Competition

Regionals (Islamabad)

2024

Pitched Solar Cleaning Automation Startup idea, with market penetration insights demonstrating strong entrepreneurial acumen and market awareness.

Technical Skills & Interests

Technical Skills: SolidWorks, ANSYS, Rapid Prototyping (3D Printing, Laser Cutting), CAD Design, Intermediate Programming (Python and C++) , Problem Identification & Solving.

Interests: Robotics, Machine Learning, Automation, Innovation, Product Design and Execution.